Image quality evaluation

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1 Introduction

Human eyes can't detect if an image is different from another. But in some cases, it can be critical to know if the image is different from another, especially in compression domain where the goal is to conserve a good quality of image with a maximum compression. In this work, we will work on the image quality evaluation using several methods.

2 Methodology

To make image quality evaluation, we will use the MSE, PSNR, SSIM and LPIPS methods between two images, original image and modified image.

2.1 MSE (Mean Square Error)

This is a simple image quality evaluation that basicaly compute the mean between the two images and take the square of it. Mathematicaly, it can be represented by formula 1 for two images I and K of size $m \times n$.

$$\text{MSE} = \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} [I(i,j) - K(i,j)]^2 \tag{1}$$

It give the sum of the square mean of each pixel, so a little value mean that the two images are very similar.

2.2 PSNR

The Peak Signal Noise Ratio (PSNR) is a measure that evaluate the quality of reconstructed images for instance after compression by comparing the maximal strength of the signal to the strength of the noise generated during reconstruction that affect the image.

Mathematicaly, it can be represented by formula 2 for two images I and K of size $m \times n$.

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{\text{MSE}} \right) \tag{2}$$

Where MAX_I^2 is the maximum value of a pixel.

As said in section 2.1, a little value of the MSE indicate that the two images are very similar. So as we have $\frac{MAX_I^2}{\text{MSE}}$, it means that when images are very similar, PSNR is high.

2.3 SSIM

The Structural Similarity Index (SSIM) is a perceptual metric used to quantify the similarity between two images, focusing on human vision. It measure the perceptual differences between two images, based on visible structure of images. So it's not a pixel-wise method, but more a method useful in term of visual perception of human.

Mathematicaly, it can be represented by formula 3 for two images I and K of size $m \times n$.

$$\text{SSIM}(I,K) = \frac{(2\mu_I\mu_K + C_1)(2\sigma_{IK} + C_2)}{(\mu_I^2 + \mu_K^2 + C_1)(\sigma_I^2 + \sigma_K^2 + C_2)} \tag{3}$$

With:

- μ_I the mean of image I
- μ_K the mean of image K
- σ_I^2 the variance of image I
- σ_K^2 the variance of image K
- σ_{IK} the covariance between image I and image K
- C_1, C_2 constant values to stabilize the division

If we suppose that image I is image K, the SSIM give us:

$$\text{SSIM}(I, I) = \frac{(2\mu_I\mu_I + C_1)(2\sigma_I^2 + C_2)}{(\mu_I^2 + \mu_I^2 + C_1)(\sigma_I^2 + \sigma_I^2 + C_2)} = \frac{(2\mu_I^2 + C_1)(2\sigma_I^2 + C_2)}{(2\mu_I^2 + C_1)(2\sigma_I^2 + C_2)} = 1$$

So a value closest to 1 indicate that the two images are perceptualy very similar.

2.4 LPIPS

3 Implementation

For the implementation, I use `python` with OpenCV to load image and to make the jpeg compression.

Else, `numpy` library is used to make operations on image such as adding a gaussian noise, compute the MSE (ref section 2.1) or the PSNR (ref section 2.2).

In my code, I work only with images in range [0, 255].

4 Results

5 Discussion

6 Conclusion